

Supplementary Dataset: Top-20 Candidates per Domain

Horizon Materials Screening — 2026-03-05 — Preprint v2 Supplement

Evidence: [A] Literature validated · [B] ML prediction · [C] Cross-domain hypothesis. Uncertainty: ±% on predicted property values. Synth. Difficulty: 1 (easy) – 5 (hard).

S1. Solid-State Electrolytes — Top 20 Candidates

#	Composition	Class	σ_{RT} (mS/cm)	E_a (eV)	ESW (V)	Mod. (GPa)	T_{syn} (°C)	Uncert.	Synth Diff.	Lit. Ref.	Evid.
1	Li3ScCl6	Halide	3.0	0.25–0.28	0.9–4.3	35	300–550	±20%	2	ACS SDL 2025	[A]
2	Ga-LLZO	Garnet	2.1	0.24–0.26	0–6.0	150	900–1000	±15%	4	RSC Garnet 2024	[A]
3	Li3InCl6	Halide	2.0	0.29–0.31	0.6–4.3	35	200–260	±20%	2	Nature 2021	[A]
4	LATP	NASICON	1.0	0.24–0.30	2.5–4.2	103	750–900	±15%	3	Solid Ion. 2023	[A]
5	Li3YBr3Cl3	Mix-Halide	1.5–3.0	0.22–0.28	2.5–4.0	30	300–500	±25%	3	JMCA 2025	[B]
6	Garnet+Halide hybrid	Composite	>5 pred	~0.22	~3–5	~80	800–1000	±30%	4	None (emergent)	[C]
7	Li6PS5Cl (argyrodite)	Sulfide	1.2–2.4	0.20–0.25	~3.0	20	300–500	±20%	2	JACS 2022	[A]
8	Li10GeP2S12 (LGPS)	Sulfide	12.0	0.18–0.22	~3.7	18	500–700	±15%	3	Nature Mat 2011	[A]
9	Al-doped LLZO	Garnet	1.6	0.26–0.30	0–6.0	140	900–1000	±15%	4	Adv Mat 2024	[A]
10	Li2ZrCl6	Halide	0.4–1.0	0.30–0.35	0.8–4.0	30	250–450	±25%	2	CM 2023	[A]
11	Li3YCl6	Halide	0.5–1.5	0.27–0.32	1.0–4.2	32	280–480	±25%	2	ACS AEM 2024	[A]
12	LLZO Ta-doped	Garnet	1.0	0.28–0.32	0–5.5	145	950–1050	±20%	4	JPS 2024	[A]
13	Li3AlF6	Fluoride	0.1–0.5	0.35–0.45	0–6.5	60	600–800	±30%	3	npj CM 2023	[B]
14	Oxyhalide Li3OCl	Oxyhalide	1.0–2.0	0.24–0.30	1.5–4.5	45	400–600	±30%	3	Nat Comm 2024	[B]
15	LIPON thin film	Amorphous	2.0	0.54	~5.5	70	RT (PVD)	±10%	4	Classic (1997)	[A]
16	Li7La3Zr2O12 (cubic)	Garnet	0.5–1.2	0.30–0.36	0–6.0	148	900–1100	±15%	4	Murugan 2007	[A]
17	Li3BO3-Li2SO4	Glass-cryst	0.1–0.8	0.40–0.50	~4.0	50	500–700	±35%	3	SSI 2022	[B]
18	Sulfide-halide hybrid	Composite	2.5–4.0	0.20–0.25	2.0–4.5	25	300–500	±30%	3	ACS SDL 2025	[C]
19	Li2MgCl4	Halide	0.3–0.8	0.32–0.40	1.0–4.5	28	200–400	±30%	2	CM 2024	[B]
20	Li3TaCl6	Halide	0.8–1.5	0.28–0.34	0.9–4.2	33	280–500	±25%	2	JMCA 2024	[A]

LGPS (rank 8) has highest reported σ but limited ESW vs Li metal — included for completeness. Ranks 6, 14, 18 are emergent hypothesis candidates.

S2. Thermoelectrics — Top 20 Candidates

#	Composition	ZT_RT	ZT_peak	T_peak (K)	κ_L (W/mK)	S (μ V/K)	T_syn (°C)	Uncert.	Synth Diff.	Lit. Ref.	Evid.
1	Ag2Se-rGO film	1.28	~1.5	300	<0.09	158	<200	±0.15	2	Nat Comm 2025	[A]
2	Na-SnSe polycryst.	~1.2	3.1	783	0.22	350	950 SPS	±0.30	3	Nat Mat 2022	[A]
3	Cu6Te2S2	~0.8	~1.8	500	0.25	180	700	±0.40	3	Adv Mat 2025	[A*]
4	GeTe multi-doped	~0.9	2.4+	700	~1.2	220	900	±0.25	3	Acta Mat 2024	[A]
5	PEDOT:PSS/BST/rGO	~0.5	1.23	415	~0.3	120	<300	±0.20	2	ACS AEM 2024	[A]
6	Yb/Ba/Ca skutterudite	~0.6	2.18	500	1.12	210	850 SPS	±0.30	4	Adv Funct 2023	[B/C]
7	SnSe/PEDOT hybrid	~1.4	~2.0	500	~0.3	300	<500	±0.40	3	None (emergent)	[C]
8	Bi2Te3 (p-type, opt.)	1.0	~1.1	340	~1.0	220	600 SPS	±0.10	3	Goldsmid 2014	[A]
9	PbTe-SrTe nanocomp.	~0.5	2.5	800	~0.5	280	750 SPS	±0.30	4	Nature 2012	[A]
10	Cu2Se superionic	~0.5	1.6	700	~0.4	130	400	±0.30	2	Nat Mat 2012	[A]
11	CoSb3 (In-filled)	~0.3	1.5	800	~0.8	200	700 SPS	±0.25	3	PRB 2019	[A]
12	Half-Heusler TaFeSb	~0.4	1.52	700	~3.0	190	900 SPS	±0.25	4	Nat Comm 2021	[A]
13	AgSbTe2	~0.5	1.7	700	~0.3	200	500	±0.35	2	APL 2004	[A]
14	Cu2ZnSnSe4 (CZTSe)	~0.3	0.9	700	~0.6	160	800	±0.30	3	JACS 2022	[A]
15	Mg3Sb2 n-type	~0.5	1.8	700	~0.6	240	800 SPS	±0.25	3	Energy ES 2019	[A]
16	InSe layered	~0.4	1.2	600	~0.9	210	600	±0.30	3	CM 2024	[B]
17	SnTe (In-doped)	~0.4	1.4	800	~0.7	180	700	±0.30	3	ACS Nano 2021	[A]
18	Oxyhalide TE hybrid	~0.3	~1.5	600	~0.5	200	500	±0.40	3	None (emergent)	[C]
19	BiSbTe/graphene hybrid	~0.7	1.1	360	~0.8	190	<300	±0.25	2	Nat Electron 2023	[A]
20	GeTe/Ag2Te nanocomp.	~0.6	2.1	700	~0.8	250	900	±0.30	4	Adv Mat 2024	[A]

* Cu6Te2S2 κ =0.25 W/mK from single study — independent replication required. Ranks 6–7, 18 are emergent/hypothesis candidates.

S3. CO2 Sorbents — Top 20 Candidates

#	Material	Class	Cap. (wt%)	Kinetics (s)	Regen E (MJ/kg)	Cycles	T_synth (°C)	Uncert.	Synth Diff.	Lit. Ref.	Evid.
1	Mg-MOF-74	MOF	27.2	~30	2.1	>1000	RT solvothermal	±5%	2	JACS 2008	[A]
2	UiO-66-NH2	MOF	24.8	~45	2.4	>800	120	±5%	2	DT 2012	[A]
3	ZIF-8+UiO hybrid	Hybrid MOF	21.0	~50	2.6	TBD	120	±15%	3	None (emergent)	[C]
4	TEPA/SBA-15	Amine-Si	18.5	~120	3.2	~500	60 (impreg.)	±8%	2	ET&F 2021	[A]
5	MIL-101(Cr)-NH2	MOF	20.3	~60	2.8	>600	150	±8%	3	CM 2012	[A]
6	HKUST-1	MOF	19.8	~40	2.5	~500	RT solvothermal	±8%	2	Science 1999	[A]
7	Zeolite 13X	Zeolite	17.5	~90	3.5	>5000	500 (calc.)	±5%	1	Classic	[A]
8	NaX zeolite	Zeolite	16.2	~100	3.6	>5000	500	±5%	1	AIChE 2019	[A]
9	MOF-177	MOF	15.8	~60	2.2	>500	RT	±8%	3	JACS 2005	[A]
10	ZIF-8	MOF	10.2	~30	2.0	>1000	85	±8%	2	JACS 2006	[A]
11	SIFSIX-3-Ni	MOF	22.0	~20	2.3	>600	85	±8%	3	Nature 2015	[A]
12	Co-MOF-74	MOF	23.1	~35	2.2	>700	RT solvothermal	±8%	2	ChemComm 2010	[A]
13	Al-MOF (MIL-53)	MOF	15.0	~80	3.0	>800	220	±8%	2	JACS 2002	[A]
14	Porous carbon amine	Carbon	16.8	~150	3.8	~300	700 (pyrolysis)	±10%	3	Carbon 2023	[B]
15	Mg(OH)2 pellets	Hydroxide	14.2	~300	4.0	>100	<200	±10%	1	Ind Eng C 2022	[A]
16	PPN-6-SO3H	PPN	17.5	~70	2.5	>500	80	±10%	3	JACS 2011	[A]
17	UiO-67	MOF	18.0	~55	2.5	>700	120	±8%	2	CrystGrowDes	[A]
18	Amine-MOF hybrid	Hybrid	25.0	~40	2.2	TBD	120	±15%	3	None (emergent)	[C]
19	Ionic liq./MOF	Composite	20.5	~35	2.0	TBD	RT	±20%	4	JMCA 2024	[B]
20	Covalent organic F	COF	19.2	~45	2.1	>400	120	±15%	3	Nat Chem 2022	[A]

Ranks 3, 18 are cross-domain hypothesis composites. Zeolites (7–8) included as established industrial baseline.

S4. Hydrogen Storage — Top 20 Candidates

#	Material	Cap. (wt%)	T_rel (°C)	Cycles	Kinetics	T_synth (°C)	Uncert.	Synth Diff.	Lit. Ref.	Evid.
1	MgH2-MOF composite	8.2	~85	>500	Good [B]	<400	±0.5 wt%	3	Pred. (emergent)	[B]
2	NaAlH4 + Ti cat.	7.5	~100	>500	Good [A]	~300	±0.3 wt%	2	Bogdanovic 1997	[A]
3	Li-amide/imide	6.5	~90	~300	Moderate	~500	±0.4 wt%	3	Nature 2002	[A]
4	MOF-177 (77K)	7.5*	~25	>500	Fast [A]	RT	±0.3 wt%	3	Rosi 2003	[A]
5	Complex hydride ML	9.1	~95	Unknown	Unknown	TBD	±1.5 wt%	5	ML prediction	[B]
6	LiBH4 nanoconf.	6.8	~200	>200	Moderate	~200	±0.5 wt%	3	JACS 2008	[A]
7	Ca(BH4)2	9.6	~170	~200	Slow [A]	~300	±0.5 wt%	3	PRB 2007	[A]
8	MgH2 + catalyst	7.6	~150	>500	Moderate	~400	±0.3 wt%	2	IJHE 2022	[A]
9	Mg(BH4)2	14.9	~290	~100	Very slow	~200	±0.5 wt%	4	Nat Chem 2010	[A]
10	NH3BH3 (thermolysis)	19.6	~70	Irrevers	Fast [A]	RT	±1.0 wt%	2	Science 2006	[A]
11	ZIF-8 H2 (77K)	5.1*	~25	>1000	Fast [A]	85	±0.3 wt%	2	JACS 2006	[A]
12	COF-108 (cryo)	21.4*	~25	>500	Fast [B]	120	±1.0 wt%	3	Science 2007	[A]
13	FeNi hollow nanop.	2.9	~50	>300	Fast [B]	200	±0.5 wt%	4	ACS Nano 2024	[B]
14	Ti-MOF composite	5.5	~100	>400	Good [B]	~300	±0.5 wt%	3	Angew 2023	[B]
15	AlH3 (alane)	10.1	~100	Irrevers	Fast [A]	Chem synth	±0.4 wt%	3	JACS 2004	[A]
16	LiNH2-MgH2	5.5	~90	>300	Moderate	~400	±0.4 wt%	3	Nat Mat 2004	[A]
17	Pd/AC H2 spillover	3.8	~25	>500	Fast [A]	~600	±0.5 wt%	3	JACS 2006	[A]
18	MgH2+MOF+catalyst	8.8	~75	>500	Good [C]	<400	±0.8 wt%	4	Emergent combo	[C]
19	NaAlH4-MOF	6.2	~100	>400	Good [B]	~300	±0.5 wt%	3	IJHE 2023	[B]
20	V-decorated graphene	3.5	~50	>500	Fast [B]	CVD	±0.5 wt%	4	Nano Lett 2022	[B]

* Cryogenic (77K) values — RT capacity significantly lower. Rank 18 is cross-domain composite hypothesis.